

Matrix functions and stability

Local Hölder continuity of the joint spectral radius

Difficulty: challenging **Importance:** interesting to the community

Status: open; Conjecture L1 in the source below

Context and notation

Let $\mathcal{H}_d$ denote the nonempty compact subsets of $\mathbb{C}^{d \times d}$. Use the spectral norm and its Hausdorff distance

$$d_H(\mathcal{M}, \mathcal{N}) = \max \left\{ \sup_{A \in \mathcal{M}} \inf_{B \in \mathcal{N}} \|A - B\|_2, \sup_{B \in \mathcal{N}} \inf_{A \in \mathcal{M}} \|A - B\|_2 \right\}.$$

The joint spectral radius is

$$\hat{\rho}(\mathcal{M}) = \lim_{k \rightarrow \infty} \max_{A_1, \dots, A_k \in \mathcal{M}} \|A_k \cdots A_1\|_2^{1/k}.$$

These definitions also apply to finite real matrix sets. The ordinary spectral radius of one matrix is written $\rho(A)$.

Problem statement

For every $d \geq 1$ and $\mathcal{M}_0 \in \mathcal{H}_d$, do there exist $r, C > 0$ such that

$$|\hat{\rho}(\mathcal{M}) - \hat{\rho}(\mathcal{N})| \leq C d_H(\mathcal{M}, \mathcal{N})^{1/d}$$

whenever $d_H(\mathcal{M}, \mathcal{M}_0) < r$ and $d_H(\mathcal{N}, \mathcal{M}_0) < r$?

Reference and status evidence

Epperlein and Wirth, [The joint spectral radius is pointwise Hölder continuous](#), Linear Algebra Appl. 704 (2025), 92–122, DOI, §2, Conjecture 3 (L1). Pointwise results in that paper do not prove this local assertion.

Additional status evidence

Searches combining “joint spectral radius” with “local Hölder”, “Lipschitz lower”, “trajectory bounds”, the authors’ names, and 2025/2026 found no later resolution. These searches supplement the explicit 2025 conjectures; they do not establish exhaustiveness. The three entries are separately named assertions in the source, not a count of dimensional cases.